# Dirt

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# What is Dirt?

**Project Goal:**

I am building Dirt, a small programming language.I am going to use it to build a turnip farming simulator game.

**Core Idea:**

I want to understand language design better by building a language and immediately using it define real application behavior. The browser app handles visuals and input, while Dirt scripts define the rules (planting, growth, harvesting, etc). Changing the Dirt script will change how the application behaves.

# Scope Control and Feasibility

This project is intentionally scoped to be realistic for a two-month timeline. Instead of creating a large or complex game, the project uses a simple turnip-farming simulation to keep the logic and UI manageable. This allows the focus to stay on language design and integration.

## Scope Decisions

- Intentionally limited to one crop
- One small grid-based farm area
- No NPCs, animals, seasons, etc
- Language features limited to essentials
- Minimal browser host application

# Risks and Success

## Risks and Mitigation Strategies

**Risk**: Language takes longer than expects

**Mitigation**: Reduce demo complexity, turnip only

**Risk**: UI polish falls behind

**Mitigation**: Focus on functionality over visuals

**Risk**: Interpreter bugs

**Mitigation**: Small test scripts and staged features

## What Success Looks Like

- Dirt scripts load and execute without crashing and burning your house down
- Changing Dirt code changes game behavior
- Basic gameplay is working (harvesting and growth)
- Clear visual feedback via popups

# Project Architecture (High Level)

**Dirt Programming Language**

This includes the lexer, parser, and interpreters, which take a Dirt script and execute it. This is the core of the project.

**Browser Host Application**

This includes visuals, user, input, and basic state like tiles and crops. It is intentionally minimal and exists mainly to give the language something concrete to control.

**Integration Layer**

This connects Dirt to the host application. The host triggers events such as interacting with a tile, and the Dirt interpreter runs the corresponding scripts.

# Dirt Language Design (In Progress)

- Block-based syntax using { }
- Event-driven scripts
- Named event blocks
- English-like keywords and functions
- Minimal punctuation
- Built-in functions for simple game actions
- Basic conditional logic (if / else)

**Example Dirt Code Snippet**

```
onInteractTile "soil" {
  if hasItem("turnipSeed") {
    plant("turnip")
    showPopup("Planted!")
  }
}
```

# Up Next

**Build the Lexer**

- Tokens: keywords, identifiers, numbers, strings, braces, parentheses, commas
- Track line/column for error messages
- Ignore whitespace

**Stop condition:** I can print a correct token list for my example scripts.

**Build the Parser -> AST**

- Parse only what I need:
  - Events blocks
  - Statements: function calls, if / else
- Produce an AST data structure

**Stop condition:** My example scripts parse into an AST without errors

# Dirt

**Project start:** Mon, 1/26/2026

**Display week:** 1

| TASK | HOURS | ASSIGNED TO | PROGRESS | START | END |
|---|---|---|---|---|---|
| **Planning and Language Design** | | | | | |
| Definition and Constraints | 2 | Ella Anderson | 100% | 1/26/26 | 1/27/26 |
| Goals and Feature Locks | 2 | Ella Anderson | 100% | 1/27/26 | 1/28/26 |
| Syntax Design | 2 | Ella Anderson | 50% | 1/28/26 | 1/30/26 |
| Example Scripts | 2 | Ella Anderson | 0% | 1/30/26 | 2/1/26 |
| **Language Implementation** | | | | | |
| Lexer Implementation | 12 | Ella Anderson | 0% | 2/1/26 | 2/7/26 |
| Parser Implementation | 14 | Ella Anderson | 0% | 2/7/26 | 2/14/26 |
| Interpreter Core and Execution Mo | 16 | Ella Anderson | 0% | 2/14/26 | 2/21/26 |
| Built-In Functions | 10 | Ella Anderson | 0% | 2/21/26 | 2/28/26 |
| **Browser-Based Host Application** | | | | | |
| Host Application Setup | 10 | Ella Anderson | 0% | 3/1/26 | 3/8/26 |
| Game Visualization | 4 | Ella Anderson | 0% | 3/8/26 | 3/15/26 |
| Player Interaction Handling | 6 | Ella Anderson | 0% | 3/15/26 | 3/22/26 |
| Popup Messages | 6 | Ella Anderson | 0% | 3/22/26 | 3/29/26 |
| **Integration and Delivery** | | | | | |
| Dirt Script in Host Application | 8 | Ella Anderson | 0% | 4/1/26 | 4/4/26 |
| Turnip Logic and Growth | 6 | Ella Anderson | 0% | 4/4/26 | 4/8/26 |
| Inventory and Harvesting | 6 | Ella Anderson | 0% | 4/9/26 | 4/12/26 |
| Documentation | 1 | Ella Anderson | 0% | 4/12/26 | 4/15/26 |
| Miscellaneous Shenanigans | 10 | Ella Anderson | 0% | 4/15/26 | 4/23/26 |
| Final Presentation | 0.5 | Ella Anderson | 0% | 4/23/26 | 4/23/26 |

